ECE 3355: Electronics

Section 18559
Fall 2019

Exam 2

Version B

November 19, 2019

Do *not* open the exam until instructed to do so. Answer the questions in the spaces provided on the question sheets and be sure to show your work. If you run out of room for an answer, continue on the back of the page. This is a closed-book/notes exam and you *may* use a calculator. You may use two crib-sheets as described in the syllabus and discussed in class. **You will have 1 hr. 15 min. to finish the exam.**

Student's Name: _____

Question	Points	Score
1	25	
2	30	
3	45	
Total:	100	

From Table 6.2:

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha}\right) e^{v_{BE}/V_T}$$

Note: For the pnp transistor, replace v_{BE} with v_{EB}.

$$i_C = \alpha i_E$$

$$i_C = \beta i_B$$

$$\beta = \frac{\alpha}{1-\alpha}$$

$$i_B = (1-\alpha)\, i_E = \frac{i_E}{\beta+1}$$

$$i_E = (\beta+1)\, i_B$$

$$\alpha = \frac{\beta}{\beta+1}$$

$V_T =$ thermal voltage $= \dfrac{kT}{q} \simeq 25\,\text{mV}$ at room temperature

Summary of Table 7.4 (Small Signal Model Parameters)

Model Parameters in Terms of DC Bias Currents

$$g_m = \frac{I_C}{V_T} \quad r_e = \frac{V_T}{I_E} = \alpha\frac{V_T}{I_C} \quad r_\pi = \frac{V_T}{I_B} = \beta\frac{V_T}{I_C} \quad r_o = \frac{|V_A|}{I_C}$$

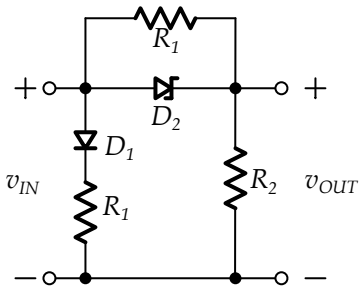
In Terms of g_m

$$r_e = \frac{\alpha}{g_m} \qquad r_\pi = \frac{\beta}{g_m}$$

In Terms of r_e

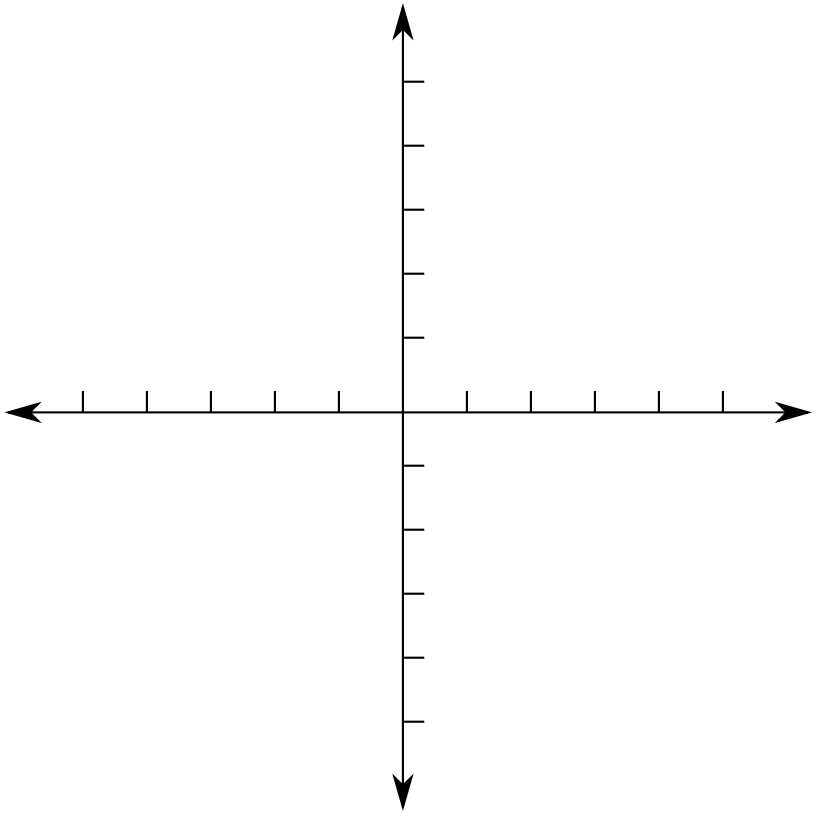
$$g_m = \frac{\alpha}{r_e} \quad r_\pi = (\beta+1)\, r_e \qquad g_m + \frac{1}{r_\pi} = \frac{1}{r_e}$$

1. (25 points) For the circuit below, assume $V_{D0} = 0.7$ volts, $V_{Z0} = 1.5$ volts, $R_1 = R_2$, and $r_D = r_Z = 0$.



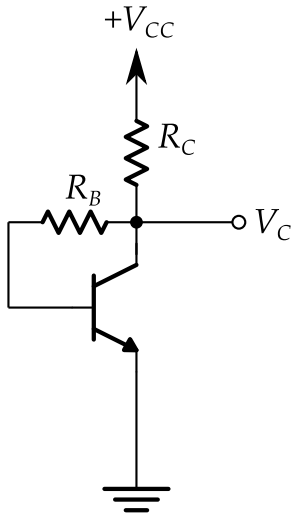
(a) Write an expression for v_{OUT} in terms of v_{IN}.

(b) Sketch v_{OUT} and v_{IN} for -4 volts $< v_{IN} < 4$ volts on the graph provided. Be sure to label the graph properly.



(c) Now, replace the Zener diode (D_1) with a regular diode, and use $V_{D0} = 0.7$ volts and $r_D = R_1 = R_2$. Write an expression for v_{OUT} in terms of v_{IN}.

2. Use the following circuit to complete this problem.

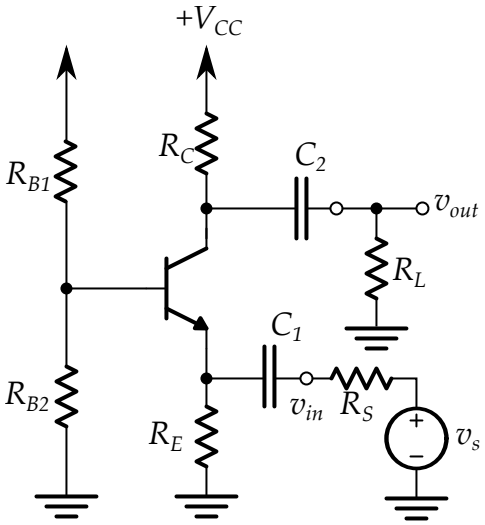


(a) (10 points) Write expressions for V_C and the currents in the transistor in terms of R_B, R_C, V_{CC}, and β.

(b) (10 points) Find values for R_B and R_C so that $V_C = V_{CC}/2$ and $r_\pi = 1\,\text{k}\Omega$ when $V_{CC} = 25\,\text{V}$ and $\beta = 100$.

(c) (10 points) Using the values for the resistors and power supplies in the previous part, what is the range of V_C when $50 \leq \beta \leq 150$?

3. Use the following circuit to complete this problem.



(a) (10 points) Assuming that the transistor is in an active mode, sketch the small signal model for the circuit.

(b) (10 points) Write an expression for the *midband gain* of the circuit, $\frac{v_{out}}{v_{in}}$, with the load and source removed. The solution should be expressed in terms of resistors, β, r_π.

(c) (10 points) Find expression for the input and output resistance for the *midband* with the source and load removed.

(d) (10 points) Write expressions for the resistances seen by the capacitors C_1 and C_2.

(e) (5 points) Make a rough sketch of a possible Bode magnitude plot for this circuit.